

# SMe Note

**Acknowledge:** Sample Space:  $\Omega$     Random Variables:  $\mathbf{Y} = (Y_1, \dots, Y_n)$     Sample Data:  $\mathbf{y} = (y_1, \dots, y_n)$      $f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_i f_{Y_i}(y_i; \theta)$ , if  $Y_i$  are independent    Distribution Function:  $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \leq y_1, \dots, Y_n \leq y_n)$   
Parameter Space:  $\Theta$     CDF:  $F_Y(y)$     PDF/PMF:  $f_Y(y)$  or  $p_Y(y)$

## 1 Basic Knowledge

**Def of expectation (Moments):** For any well-behaved function  $h$  on  $\mathbb{R}$ .  $\mathbb{E}[h(Y)] := \sum_i h(y_i) \mathbb{P}(Y = y_i)$      $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y) f_Y(y) dy$

**Def of Model:** A collection of distributions indexed by a parameter  $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \in \Theta\}$  is model.

**Likelihood Function:**  $L(\theta; \mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$     **MLE:**  $\hat{\theta}(\mathbf{y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{y})] = [U(\hat{\theta}) = 0]$  ps: MLE has invariant property

**(Strong) Likelihood Principle:**  $L(\theta; x) \propto L(\theta; y)$  [Strong:  $L_f(\theta; x) \propto L_g(\theta; z)$ ;  $f, g$  are statistical model]  $\Rightarrow$  Same Inferential Conclusions.

**Def of Identifiability:** A model is identifiable if  $f(y; \theta_1) = f(y; \theta_2), \forall y$  then  $\theta_1 = \theta_2$  (i.e.  $\text{map} : \theta \rightarrow f(y; \theta)$  is 1-1)

• Method: 1.By def  $\Rightarrow$  ratio not depend on  $y \Rightarrow \theta_1, \theta_2$  relationship. (with any additional constraint)    2.MLE not unique  $\Rightarrow$  not identifiable    3.Give an example  $\theta$     4.Fisher information

## 2 Properties of the Likelihood and the MLE

**Positive [Semi-]definite matrix M:**  $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T \mathbf{M} \mathbf{z} > 0$  [ $\geq 0$ ]  $\Leftrightarrow$  ( $> 0$ ) <sup>1</sup> n positive eigenvalues. <sup>2</sup> full rank. <sup>3</sup> invertible

|                                                                            |                                                                                                                                                                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                     |                          |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Score Func<br>/ Vector U                                                   | Single: $U(\theta, \mathbf{Y}) = \ell'(\theta; \mathbf{Y})$                                                                                                                  | $\mathbb{E}[U] = \mathbf{0}$                                                                                                                                       | $Var[U] = \mathbf{I}(\theta)$                                                                                                                                                                                                                                                       |                          |
|                                                                            | Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1} \ell(\theta; \mathbf{Y}), \dots, \partial_{\theta_p} \ell(\theta; \mathbf{Y})]^T \in \mathbb{R}^p$                      | $\mathbb{E}[\mathbf{U}] = \mathbf{0}$                                                                                                                              | $Var[\mathbf{U}] = \mathbf{I}(\theta) \quad I_{j,k} = Cov(\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j}, \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}) = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ |                          |
| CLT of U                                                                   | Single: $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, \mathbf{I}(\theta_0)), n \rightarrow \infty$                                                                            |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                     |                          |
|                                                                            | Multiple: $\mathbf{U}(\theta_0; \mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I}(\theta_0)), n \rightarrow \infty$                                                      |                                                                                                                                                                    | Also: $\mathbf{U}(\theta_0; \mathbf{Y})^T \mathbf{I}^{-1}(\theta_0) \mathbf{U}(\theta_0; \mathbf{Y}) \sim \chi_p^2$                                                                                                                                                                 |                          |
| Fisher<br>Information I                                                    | Single: Observed: $J(\theta, \mathbf{y}) = -U'(\theta, \mathbf{y}) = -\ell''(\theta; \mathbf{y})$                                                                            | Expected: $\mathbf{I}(\theta) = -\mathbb{E}[\ell''(\theta; \mathbf{Y})]$                                                                                           | Unit: $i(\theta) = \mathbf{I}(\theta)/n = -\mathbb{E}[\ell''(\theta; Y_i)]$                                                                                                                                                                                                         |                          |
|                                                                            | Multiple: Observed: $\mathbf{J}(\theta; \mathbf{y}) \in \mathbb{R}^{p \times p}, J_{j,k} = -\frac{\partial^2 \ell(\theta, \mathbf{y})}{\partial \theta_j \partial \theta_k}$ | Expected: $\mathbf{I}(\theta) \in \mathbb{R}^{p \times p}, I_{j,k} = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ | ps: $\mathbf{I}(\theta)$ always positive semi-definite                                                                                                                                                                                                                              |                          |
| $\mathbf{I} \uparrow \Rightarrow \mathbf{Y}$ 对 $\theta$ 越敏感; 信息量大; 参数估计更精确 |                                                                                                                                                                              | A model is <i>identifiable</i> iff $\mathbf{I}(\theta)$ is positive definite. ps:(一维) identifiable if $\mathbf{I}(\theta) \neq \mathbf{0}$                         |                                                                                                                                                                                                                                                                                     |                          |
| CLT of MLE<br>CLT of I                                                     | Single: $\mathbf{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, \mathbf{I}^{-1}(\theta_0)), n \rightarrow \infty$                                 | $\widehat{Var}(\hat{\theta}) \approx \mathbf{I}^{-1}(\hat{\theta})$                                                                                                | $ESE(\hat{\theta}) \approx \sqrt{\mathbf{I}^{-1}(\hat{\theta})}$                                                                                                                                                                                                                    | estimated standard error |
|                                                                            | Multiple: $\mathbf{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}_p(\theta_0, \mathbf{I}^{-1}(\theta_0)), n \rightarrow \infty$                             | Also: $(\hat{\theta} - \theta_0)^T \mathbf{I}(\theta_0) (\hat{\theta} - \theta_0) \sim \chi_p^2$                                                                   |                                                                                                                                                                                                                                                                                     |                          |

### 2.1 For Single Parameter

\*Random Variable / Function:  $L(\theta; \mathbf{Y})$  is random function (NOT pdf). For fixed  $\theta$ , it's r.v.     $\hat{\theta}(\mathbf{Y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{Y})]$  is r.v.    ps:  $\mathbf{I}^{-1}, \mathbf{J}^{-1}$  是指其负一次方, 不是反函数    True Parameter:  $\theta_0$

| Name                                                                                                                                                                                                                                                                                                                 | Test Statistic                                                                                                                                                                                    | Hypothesis                                        | CI                                                                                        | Other                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score Test                                                                                                                                                                                                                                                                                                           | Given $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, \mathbf{I}(\theta_0))$<br>$T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{\mathbf{I}(\theta_0)}} \sim \mathcal{N}(0, 1)$                     | $H_0 : \theta = \theta_0$<br>不用 MLE, 泰勒展开         | $\left\{ \theta : \frac{U(\theta)}{\sqrt{\mathbf{I}(\theta)}} < z_{\alpha, 1}^2 \right\}$ | or $T_{Score}^2(\mathbf{Y}) \sim \chi_1^2$<br>$T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{\mathbf{I}(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{\mathbf{J}(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{\mathbf{J}(\hat{\theta})}}$                                                                                                   |
| Wald Test or<br>Maximum Likelihood(ML) Test                                                                                                                                                                                                                                                                          | $T_{Wald}(\mathbf{Y}) = \sqrt{\mathbf{I}(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta} - \theta_0}{\sqrt{\mathbf{I}^{-1}(\theta_0)}}$<br>$T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0, 1)$ | $H_0 : \theta = \theta_0$<br>构建 CI                | $\{ \theta : (\hat{\theta} - \theta)^2 \mathbf{I}(\theta) < z_{\alpha, 1}^2 \}$           | or $T_{Wald}^2 \sim \chi_1^2$ ; $n \rightarrow \infty T_{Wald} \simeq T_{Score}$<br>$T_{Wald}(\mathbf{Y}) \approx \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{I}^{-1}(\hat{\theta})}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{J}^{-1}(\theta_0)}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{J}^{-1}(\hat{\theta})}}$ |
| log-Likelihood<br>Ratio Test                                                                                                                                                                                                                                                                                         | $T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi_1^2$ as $n \rightarrow \infty$<br>$LR = L(\theta_0)/L(\hat{\theta}), T_{LR} = -2\ln(LR)$                                   | $H_0 : \theta = \theta_0$<br>最小 CI, Most Powerful | $\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, 1}^2 \}$                   | ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \simeq (\theta_0 - \hat{\theta})^2 J(\hat{\theta})$                                                                                                                                                                                                                                       |
| $T_{Score} \simeq T_{Wald} \simeq T_{LR}$ as $n \rightarrow \infty$ ; $J(\theta_0) \approx \mathbf{I}(\theta_0)$ as $n \rightarrow \infty$ ; When $\ell(\theta)$ 是关于 $\theta$ 的二次函数 $\Rightarrow$ All same.    Geometric: ( $T_S$ , 比斜率), ( $T_W$ , 比宽度), ( $T_{LR}$ , 比高度) ps: $z_{\alpha, n} = \chi_{\alpha, n}^2$ |                                                                                                                                                                                                   |                                                   |                                                                                           |                                                                                                                                                                                                                                                                                                                                       |

### 2.2 For Single Parameter (Numerical Method)

|                                           |                                                                                                                                                                                           |                                                                                                                                    |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suppose initial value solution $\theta_0$ | Newton-Raphson Method    ps: also called Newton's method.                                                                                                                                 | Fisher's Method of Scoring    ps: also called scoring method                                                                       |
| Formula                                   | $\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$                                                                             | $\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{\mathbf{I}(\theta_r)}$ |
| Advantages                                | Converge <i>very quickly</i> near optimum                                                                                                                                                 | $\mathbf{I}$ is constant / easier to calculate; Converge for wider $\theta_0$                                                      |
| Stopping Rule                             | Given tolerance level $\varepsilon$ : <sup>1</sup> Stop after $r$ if $ U(\theta_r)  \leq \varepsilon$ (Common)    or <sup>2</sup> Stop after $ \theta_r - \theta_{r-1}  \leq \varepsilon$ |                                                                                                                                    |

### 2.3 For Multiple Parameters

\*此节中,  $\theta$  代表向量,  $\theta = (\theta_1, \dots, \theta_p)$      $\mathbf{I}^{-1}$  指其逆矩阵    True Parameter:  $\theta_0$

| Name                                                                                                                                                                                                                              | Test Statistic ( $T_{Score}^2, T_{Wald}^2, T_{LR}$ )                                         | Hypothesis          | (Confidence Region) CR                                                                                       | <b>Marginal Normal</b> distribution $\hat{\theta}$ :<br>$\hat{\theta}_j \sim \mathcal{N}(\theta_j, [I^{-1}(\underline{\theta}_0)]_{jj})$ $\theta_j$ : true para<br>· $ESE(\hat{\theta}_j) = \sqrt{[I^{-1}(\hat{\theta})]_{jj}}$<br>· $CI(\theta_j): \hat{\theta}_j \pm z_{\alpha} ESE(\hat{\theta}_j)$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score Test                                                                                                                                                                                                                        | $\mathbf{U}(\theta_0)^T \mathbf{I}^{-1}(\theta_0) \mathbf{U}(\theta_0) > z_{\alpha, p}$      | $\theta = \theta_0$ | $\{ \theta : \mathbf{U}(\hat{\theta})^T \mathbf{I}^{-1}(\theta) \mathbf{U}(\hat{\theta}) < z_{\alpha, p} \}$ |                                                                                                                                                                                                                                                                                                        |
| Wald Test                                                                                                                                                                                                                         | $(\hat{\theta} - \theta_0)^T \mathbf{I}(\theta_0) (\hat{\theta} - \theta_0) > z_{\alpha, p}$ | $\theta = \theta_0$ | $\{ \theta : (\hat{\theta} - \theta)^T \mathbf{I}(\theta) (\hat{\theta} - \theta) < z_{\alpha, p} \}$        |                                                                                                                                                                                                                                                                                                        |
| log-Likelihood Ratio Test                                                                                                                                                                                                         | $2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha, p}$                                     | $\theta = \theta_0$ | $\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, p} \}$                                        |                                                                                                                                                                                                                                                                                                        |
| $\mathbf{I}(\theta_0) \approx \mathbf{I}(\hat{\theta}), \mathbf{J}(\theta_0), \mathbf{J}(\hat{\theta})$ $ps : z_{\alpha, p} = \chi_{\alpha, p}^2$ $T_{LR}$ : simple hypothesis testing. ( $\dim(\mathcal{A}) = 0$ ) $\theta$ 都是向量 |                                                                                              |                     |                                                                                                              |                                                                                                                                                                                                                                                                                                        |

**Composite Hypothesis Test:**  $H_0 : \underline{\theta} \in \mathcal{A}, H_1 : \underline{\theta} \notin \mathcal{A}$   $\mathcal{A}$ : reduced model     $LR = \frac{\max_{\theta \in \mathcal{A}} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\underline{\theta}_{\mathcal{A}})}{L(\underline{\theta}_{\Theta})}$     ps: 多维未知参数  $H_0$  只想其中几个时用 ||  $\dim(\mathcal{A})$  常数时为 0

· **Generalised Likelihood Ratio Test:**  $T_{GLR} = -2\ln(LR) = 2(\ell(\hat{\theta}_{\mathcal{A}}) - \ell(\hat{\theta}_{\Theta})) \sim \chi_{p-r}^2$      $p : \#$  of parameters.     $r = \dim(\mathcal{A})$

### 2.4 For Multiple Parameter (Numerical Method)

| Method (start at $\theta_0$ )                                                                                                                                    | Newton-Raphson Method                                                                                                                  | Fisher's Method of Scoring                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formula                                                                                                                                                          | $\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r) \mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r) \mathbf{U}(\theta_r)$ | $\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1} \mathbf{U}(\theta_r) = \theta_r + \mathbf{I}^{-1}(\theta_r) \mathbf{U}(\theta_r)$ |
| ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$    If $\mathbf{I} = \text{diag}(a, b, \dots) \Rightarrow \mathbf{I}^{-1} = \text{diag}(a^{-1}, b^{-1}, \dots)$ |                                                                                                                                        |                                                                                                                                                      |

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

**Basic Idea:** Consider  $\theta$  as a r.v.. Bayesian approach  $\Rightarrow$  Probability are used to represent degree of believe about the unknown  $\theta$

**Prior Distribution  $p(\theta)$ :** Prior distribution  $p(\theta)$  contains all available prior information about  $\theta$  before observing any data.

· **Hyper-Parameters:** 由于  $p(\theta)$  是已知, 服从某种分布 (with parameter  $\gamma$ ). We denoted  $p(\theta)$  as  $p(\theta; \gamma)$  where  $\gamma$  is Hyper-parameters. 定义:  $\gamma_{prior} \rightarrow \gamma_{post}$

**Likelihood function  $p(y|\theta) = f(y|\theta)$ :** Any observed data is considered as a realisation of r.v.  $Y$  with  $f(y|\theta)$

**Posterior Distribution  $p(\theta|y)$ :** Posterior distribution  $p(\theta|y)$ , contains all available knowledge.

· **Bayes' Rule / Theorem:**  $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{f(y|\theta)p(\theta)}{m(y)}$      $p(y) = m(y) = \int_{\theta \in \Theta} f(y|\theta)p(\theta)d\theta$  (Normalizing constant)

|             |                                                                                        |                                                                                                                           |                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Name        | MAP <sup>Maximum A Posteriori</sup>                                                    | MMSE <sup>Posterior Mean / Bayesian Minimum Mean Squared Error Estimator</sup>                                            | Posterior Median $\hat{\theta}_{median}$                                                                                             |
| Single Para | $\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta y)$                                  | $\hat{\theta}_{MMSE} = E[\theta y] = \int_{\theta \in \Theta} \theta p(\theta y)d\theta$                                  | $\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta y)d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta y)d\theta$ |
| Name        | Posterior Var                                                                          | $C_{\alpha} = [a, b]$ (1 - $\alpha$ )100% Bayesian Credible Intervals (or Simply Credible Intervals)                      |                                                                                                                                      |
| Single Para | $Var[\theta y] = \int_{\theta \in \Theta} (\theta - E[\theta y])^2 p(\theta y)d\theta$ | $P(a \leq \theta \leq b y) = \int_a^b p(\theta y)d\theta = 1 - \alpha$ ps: $P(\theta < a y) = P(\theta > b y) = \alpha/2$ |                                                                                                                                      |

3.2 Exact Bayesian Inference with Conjugate priors

**Conjugate Prior:** For  $\theta$ ,  $p(\theta)$  is Conjugate Prior for  $p(y|\theta) = f(y|\theta)$  iff for  $p(\theta)$  用 Bayes' Rule 得到的  $p(y|\theta)$  都在同一类型的分布

**Conjugate Family:** The collection of all Conjugate Prior.

**Choice of Prior:** Observed data number  $n$ : <sup>1</sup>  $n$  small  $\Rightarrow$  Posterior shows a great dependency of the choice of prior.

<sup>2</sup>  $n$  middle  $\Rightarrow$  posterior less affected by choice of prior. <sup>3</sup>  $n$  large  $\Rightarrow$  choice of prior has limited effect on posterior. (Likelihood has a strong impact)

**Comparing the frequentist and Bayesian approaches:** <sup>1</sup>  $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$  for most cases, especially  $n$  large. <sup>2</sup>  $n$  large:  $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}$ ,  $CI \approx C_{\alpha}$

4 Distribution Table

| Distribution                                                                                                                                        | PDF                                                                                                                                                              | Mean                                                                                                | Var                                                    | Other                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bernoulli</b> $X \sim Bern(p)$                                                                                                                   | $p^x(1-p)^{1-x}, x \in \{0, 1\}$                                                                                                                                 | $p$                                                                                                 | $p(1-p)$                                               | If $X_i$ are i.i.d. $\sum X_i \sim Bin(n, p)$                                                                                                    |
| <b>Binomial</b> $X \sim Bin(n, p)$                                                                                                                  | $\binom{n}{x} p^x (1-p)^{n-x}$                                                                                                                                   | $np$                                                                                                | $np(1-p)$                                              | $X, Y \sim Bin(n, p), Bin(m, p)$<br>$X + Y \sim Bin(n + m, p)$                                                                                   |
| For $X \sim Bin(n, p), n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$ , then $X \sim Poission(\lambda)$                             |                                                                                                                                                                  | For $X \sim Bin(n, p), n \rightarrow \infty$ and pdf symmetric, $\Rightarrow X \sim N(np, np(1-p))$ |                                                        |                                                                                                                                                  |
| <b>Geometric</b> $X \sim Geom(p)$                                                                                                                   | $(1-p)^{x-1}p$                                                                                                                                                   | $1/p$                                                                                               | $(1-p)/p^2$                                            | $E[X^2] = 2/p^2$                                                                                                                                 |
| <b>Negative Binomial</b> $X \sim NB(r, p)$                                                                                                          | $\binom{r+x-1}{x} (1-p)^r p^x, r \geq 1$                                                                                                                         | $\frac{rp}{1-p}$                                                                                    | $\frac{rp}{(1-p)^2}$                                   | 或者 $p = 1 - q$ , 注意!                                                                                                                             |
| <b>Poission</b> $X \sim Po(\lambda)$                                                                                                                | $e^{-\lambda} \cdot \frac{\lambda^x}{x!}$                                                                                                                        | $\lambda$                                                                                           | $\lambda$                                              | $E[X^2] = \lambda^2 + \lambda$                                                                                                                   |
| For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$ , then $X + Y \sim Po(\lambda_1 + \lambda_2)$                                                          |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| <b>Multinomial</b> $x_1 + \dots + x_k = n$<br>$(X_1, ..., X_k) \sim MN(n, p_1, ..., p_k)$                                                           | $\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, 0 < p_i < 1$                                                                                              | $np_i$                                                                                              | $np_i(1-p_i)$                                          |                                                                                                                                                  |
| <b>Uniform</b> $X \sim \mathcal{U}([a, b])$                                                                                                         | $\frac{1}{b-a}$                                                                                                                                                  | $\frac{a+b}{2}$                                                                                     | $\frac{(b-a)^2}{12}$                                   | $E[X^2] = \frac{b^2+ab+a^2}{3}$                                                                                                                  |
| <b>Normal</b> $X \sim \mathcal{N}(\mu, \sigma^2)$                                                                                                   | $\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$                                                                                           | $\mu$                                                                                               | $\sigma^2$                                             | $Y \sim N_k(\mu, \Sigma), \mu$ vector<br>$f_Y(y_1, ..., y_k) = \frac{e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1}(y-\mu)}}{\sqrt{(2\pi)^k det(\Sigma)}}$ |
| For $X, Y \sim N(\mu_1, \sigma_1^2), Po(\mu_2, \sigma_2^2)$ , then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$ |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| <b>Exponential</b> $X \sim Exp(\lambda)$                                                                                                            | $\lambda e^{-\lambda x}$                                                                                                                                         | $\frac{1}{\lambda}$                                                                                 | $\frac{1}{\lambda^2}$                                  | $E[X^n] = \frac{n!}{\lambda^n}$                                                                                                                  |
| <b>Beta</b><br>$X \sim Beta(\alpha, \beta), \alpha, \beta > 0$                                                                                      | $\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, x \in [0, 1]$                                                                                             | $\frac{\alpha}{\alpha+\beta}$                                                                       | $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ | $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$                                                                    |
| When $Beta(\alpha, \beta)$ has maximum probability, $x = (\alpha - 1)/(\alpha + \beta - 2)$                                                         |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| <b>Gamma</b><br>$X \sim \Gamma(\alpha, \beta), \alpha, \beta > 0$                                                                                   | $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$                                                                                        | $\frac{\alpha}{\beta}$                                                                              | $\frac{\alpha}{\beta^2}$                               | $\Gamma(\alpha) = (\alpha - 1)!$<br>$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$                                                      |
| <b>Student's t</b><br>$X \sim t_v, v > 0$                                                                                                           | $\frac{1}{\sqrt{v}B(\frac{v}{2}, \frac{1}{2})} (1 + \frac{x^2}{v})^{-\frac{v+1}{2}}$                                                                             | 0                                                                                                   | $\frac{v}{v-2}$                                        | If $Z \sim N(0, 1), Y \sim \chi_n^2$<br>$\frac{Z}{\sqrt{Y/n}} \sim t_n$                                                                          |
| PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution)                                                  |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| When $n \rightarrow \infty$ it will converges to $N(0, 1)$ . <b>Cauchy Distribution:</b> $X \sim t_1$ and its mean and var are undefined.           |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| <b>Chi-Squared</b><br>$X \sim \chi_v^2, v > 0$                                                                                                      | $\frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$                                                                     | $v$                                                                                                 | $2v$                                                   | -                                                                                                                                                |
| If $X_i \sim N(0, 1)$ , then $\sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$ , then $X + Y \sim \chi_{n+m}^2$                      |                                                                                                                                                                  |                                                                                                     |                                                        |                                                                                                                                                  |
| <b>F</b> $X \sim F(\lambda, v), v > 0$<br>ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$                                                                 | $\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2}, \frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$ | $\frac{v}{v-2}$                                                                                     | $\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$ | $Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$<br>$X \sim F_{n_1, n_2}, X^{-1} \sim F_{n_2, n_1}$                       |

5 Review

Law of Total Probability:  $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$  ,  $\cup_n B_n = \Omega$  ,  $B_n$  disjoint

|      |                                   |                                         |                    |                               |                               |
|------|-----------------------------------|-----------------------------------------|--------------------|-------------------------------|-------------------------------|
| Type | Type I error                      | Type II error                           | -                  | $H_0$ is T                    | $H_0$ is F                    |
| Def  | $\mathbb{P}(Rej\ H_0 H_0\ is\ T)$ | $\mathbb{P}(Fail\ Rej\ H_0 H_0\ is\ F)$ | Fail to $Rej\ H_0$ | Right                         | Type II ( $\beta$ ) 与 size 有关 |
|      |                                   |                                         | $Rej\ H_0$         | Type I ( $\alpha$ ) 与 size 无关 | Right (Power)                 |

Significance Level:  $\alpha = \mathbb{P}(Rej\ H_0|H_0\ is\ T)$

P-value: 1.One-side:  $p = \mathbb{P}(T \geq t; T \sim H_0)$                       2.Two-sides:  $p = 2 \min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test:  $\mathbb{P}(Rej\ H_0|H_0\ is\ F) = 1 - \beta = \mathbb{P}(Z \in C|\mu = \mu^*)$  · 1.Sample size  $\uparrow$ , Power  $\uparrow$ ;    2. $\sigma^2 \downarrow$ , Power  $\uparrow$ ;    3.Effect size  $|\theta_0 - \theta^*| \uparrow$ , Power  $\uparrow$ ;    4. $\alpha \uparrow$ , Power  $\uparrow$ .

Def of Confidence Interval (CI):  $100(1 - \alpha)\%$  Confidence Interval for  $\theta$  is  $[l(X), u(X)]$ ,  $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$ .

|                             |                                                                        |                                                      |          |                                                                                                                                   |
|-----------------------------|------------------------------------------------------------------------|------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| (Convergence of r.v.) Name  | Definition                                                             | written as                                           | Example  |                                                                                                                                   |
| Convergence in Probability  | If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$ .             | $X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$ | CLT 最弱:  | 见下面                                                                                                                               |
| Convergence in Distribution | If $\forall \epsilon > 0, \lim \mathbb{P}( X_n - X  > \epsilon) = 0$ . | $X_n \xrightarrow{p} X$                              | WLLN 中等: | If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ .<br>$\Rightarrow \bar{X}_n \xrightarrow{p} \mu$                      |
| Almost Sure Convergence     | If $\mathbb{P}(\lim X_n = X) = 1$                                      | $X_n \xrightarrow{a.s.} X$                           | SLLN 最强: | If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ ,<br>$Var[X_i] < \infty \Rightarrow \bar{X}_n \xrightarrow{a.s.} \mu$ |

Central Limit Theorem (CLT): If  $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$  and  $Var[X_i] < \infty$ .  $\Rightarrow \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$     样本均值的分布会趋近于正态分布

6 Writting Format

**Distribution:** 产生新 distribution 的一种方法: 通过条件概率来.     $\mathbb{P}(A|B) = \mathbb{P}[(A \cap B)/B]$   
**p-value:** p-value measures the plausibility that the data were generated under  $H_0$ .    **Low p-value:** low p-value suggests  $H_0$  is incompatible with the data and hence likely to be false.  
**CI:** The set of values (e.g. $\theta$ ) are not rejected at significance level  $\alpha$  under  $H_0$ .    If data are generated under  $H_0$ , there is a probability  $(1 - \alpha)$  for the true  $\theta_0$  to be within the  $(1 - \alpha)100\%$  CI.  
**Linear Regression Assumptions:** Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.  
**Hypothesis Testing:**    **Steps:** <sup>1</sup> State  $H_0, H_1$ ; <sup>2</sup> Select Test Statistic  $T$ ; <sup>3</sup> distribution of  $T$  under  $H_0$ ; <sup>4</sup> observed test statistic; <sup>5</sup> critical value / region, p-value, or CI; <sup>6</sup> conclusion.    **Reject  $H_0$ :** We would reject  $H_0$  in favour of the  $H_1$ . (So that ...)    **Fail to reject  $H_0$ :** We fail to reject  $H_0$ . Hence, there is strong evidence against the null hypothesis.  
**MLE: Using (Compare) Newton's Method = ( $\neq$ ) Scoring Method:** Because the second derivative of likelihood function doesn't (does) include the r.v.  
**MLE Inference Parameter** $|\mathbb{P}(\theta < ?) = \alpha$  is false: We cannot make such probability statement.  $\{\theta\}$  is a fixed value (but unknown).  
**Geometric Comparison of  $T_{Score}, T_{Wald}, T_{LR}$ :** <sup>1</sup> $T_{Score}$  (Slope Comparison): Represents the slope of the likelihood function at  $\theta_0$ ; A steeper slope indicates that the likelihood function is changing more rapidly at  $\theta_0$ , which implies a greater difference between  $\hat{\theta}$  and  $\theta_0$ .    <sup>2</sup> $T_{Wald}$  (Width Comparison): Compares the difference between  $\hat{\theta}$  and  $\theta_0$ .    <sup>3</sup> $T_{LR}$  (Height Comparison): Compares the maximum likelihood value  $L(\hat{\theta})$  and the likelihood value at  $\theta_0, L(\theta_0)$ . A larger height difference indicates a greater discrepancy between  $\hat{\theta}$  and  $\theta_0$ .    **Other** 部分公式可以近似的理由: This is due to  $\hat{\theta}$  being a consistent estimator of  $\theta_0$ , and  $J(\theta_0)$  being a consistent estimator of  $I(\theta_0)$